GLA UNIVERSITY, MATHURA
Department of Computer Engineering Application

SYNOPSIS
for B.Tech CSE 3rd Year

Project Title: E-Commerce Platform

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Abstract

This project aims to develop a full-stack E-commerce platform that enables users to browse, search, and purchase products online efficiently. The system includes authentication, product management, shopping cart, payment integration, and order tracking. It will be developed using the MERN stack to ensure scalability and performance.

Keywords

E-commerce, MERN Stack, MongoDB, Express.js, React.js, Node.js, Web Application

1. Introduction

E-commerce platforms provide convenience and accessibility for online shopping. This project focuses on developing a scalable and user-friendly platform with admin control and secure transactions.

2. Problem Statement

Traditional shopping systems are time-consuming and limited. Existing platforms may lack usability and security. A robust and efficient system is required.

3. Objectives

- Develop a responsive website
- Implement authentication
- Enable product browsing
- Implement shopping cart
- Ensure secure payments
- Provide admin control

4. Literature Review

Existing systems like Amazon demonstrate scalability and usability. Research highlights MERN stack advantages and importance of secure systems.

5. Proposed Methodology

The system uses MERN stack with React frontend, Node/Express backend, and MongoDB database. Development follows modular approach.

6. System Architecture

Client-server architecture with React UI, Node.js APIs, and MongoDB database. Includes modules like user, product, cart, and admin.

7. Implementation Plan

Planning, design, backend development, frontend development, integration, testing, and deployment.

8. Expected Results

A fully functional E-commerce application with secure transactions, efficient management, and responsive UI.

9. Conclusion

The system enhances online shopping experience and demonstrates full-stack development capabilities.

References

1. Laudon, E-commerce, Pearson, 2020
2. MongoDB Guide, O'Reilly, 2019
3. React Docs
4. Node.js Docs